Imputing diffusion profiles of white matter fiber tracts in early brain development:

an extensive evaluation of normative, local, graph and transformer models

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Abstract

Along-tract diffusion profiles are a common summary of white matter microstructure, but in infant and toddler cohorts many profile values are missing: quality control rejects tracts that run through a regional artifact, tracts are truncated at their ends, and a whole model fit (e.g. NODDI) can be unavailable for a scan. Analyses that require complete data either drop these scans or fill the gaps ad hoc. We report an extensive evaluation of imputation methods for such profiles, carried out while developing the `impute-profiles` command of the open-source DTIPlayground package. On 423 scans of 208 children aged 0–65 months from the Baby Connectome Project (69 atlas tracts, 5,388 profile locations, FA, MD, RD, AD, NDI, ODI and FWF), we hid measured values in five patterns (interior gaps, end gaps, whole profiles, all tracts through a spherical region, and a whole family of measures) and compared several dozen settings of six method families on identical hidden values, with 3 random draws of 54 validation scans and folds by subject. Imputation from a scan's own values beat the age-bin mean only for gaps and missing ends along a profile; whole profiles and regions were 1.8 to 2.4 times as far off. Imputing deviations from the age-bin norm made every pattern better than the norm (overall RMSE 0.688 of the age-bin mean's). A population model (probabilistic PCA of all tracts and measures) with a sine-activated network (SIREN) imputing its residuals within 4 mm reached 0.614, and could impute a missing measure family (0.696 for the tensor measures, 0.798 for NODDI). Amortized networks that read a scan's observed values directly (a graph U-Net and a transformer, both correcting the population prediction) were worse alone (0.673, 0.632) but much better for a missing family (transformer 0.425 and 0.623); blending them with the population model plus SIREN, with weights per measure and distance to the nearest observed value, gave 0.569 (paired per-scan ratio 0.923 against population model plus SIREN, better on 99% of scan draws), and 0.540 when the transformer was also trained on 2,293 scans of the HBCD study. Functa, a partial VAE, age-dependent PCA, acquisition correction, context inputs and seed ensembles did not help, and pooling the second cohort into the population model made it worse. Errors were lowest for MD, RD and NDI and highest for ODI, FWF and FA, and grew with the distance to the nearest observed value. We give practical recommendations for imputing tract profiles in developmental cohorts.

Keywords: diffusion MRI; tractometry; fiber tract profiles; missing data; imputation; normative modeling; infant brain development; transformer; graph neural network; probabilistic PCA

1 Introduction

Along-tract profiles sample diffusion measures such as fractional anisotropy (FA) or mean diffusivity (MD) at regular arc-length positions along a fiber tract, and compare those profiles across

subjects (Goodlett et al., 2009; Colby et al., 2012; Yeatman et al., 2012; Verde et al., 2014). They are widely used in studies of white matter development, where the measures change fast and non-linearly over the first years of life (Geng et al., 2012; Dubois et al., 2014), and they increasingly include multi-compartment measures such as the neurite density (NDI), orientation dispersion (ODI) and isotropic free water fraction (FWF) of NODDI (Zhang et al., 2012). Infant and toddler data are, however, incomplete: motion, susceptibility and signal dropout corrupt local regions; quality control (QC) then rejects all tracts that pass through such a region; tracts are truncated where the tractography or the atlas mapping misses their ends; and some scans lack a model fit altogether. Many statistical tools for profiles (functional data analysis, mixed models over locations, multivariate and machine-learning analyses of all tracts jointly, Chamberland et al., 2019; Richie-Halford et al., 2021) assume complete profiles, so that a scan missing a few profile values is either discarded in full or patched by ad-hoc interpolation.

Imputation of missing values is a well studied problem (Rubin, 1976; Little and Rubin, 2019), with general methods ranging from nearest-neighbour (Troyanskaya et al., 2001) and chained-equation (van Buuren and Groothuis-Oudshoorn, 2011) imputation to deep generative models (Mattei and Frellsen, 2019; Ma et al., 2019). Tract profiles have structure that generic methods do not exploit: values vary smoothly along a tract; neighbouring tracts are dense in the brain; the measures of one scan are strongly coupled; and, in a developmental cohort, the expected value at each location depends strongly on age. It is not obvious which of these sources of information matters for which kind of missing value, how much a normative reference adds, and whether modern neural architectures (implicit neural representations, graph networks, transformers) bring anything beyond a well-specified linear population model.

This paper reports an extensive empirical evaluation of imputation methods for tract profiles in a normative infant and toddler cohort, the Baby Connectome Project (BCP; Howell et al., 2019). The evaluation was carried out in six phases while developing the `impute-profiles` command of the open-source DTIPlayground package (NIRAL, 2026), and every phase set or confirmed the defaults of the released tool. The emphasis of this paper is not methodological novelty but the knowledge the evaluation yields. We ask: (i) which approaches work for which missingness patterns (interior gaps, end gaps, whole profiles, all tracts through a region, a whole family of measures); (ii) how much a normative model helps; (iii) what population models, local three-dimensional (3-D) methods, graph networks and transformers each contribute; (iv) what did not help (functa, a partial variational autoencoder, age-dependent population models, acquisition correction, extra context inputs, seed ensembles); (v) whether pooling a second, larger cohort (the HEALthy Brain and Child Development study, HBCD; Volkow et al., 2021; Dean et al., 2024) helps, and how it should be done; (vi) how the error varies by measure, age and pattern; and (vii) what practitioners should do. All methods were compared on identical hidden values, with three random draws of the validation scans, folds by subject, and paired per-scan statistics.

2 Related work

Tract profiles. Along-tract analysis was introduced for DTI by sampling diffusion tensor measures along atlas or subject fiber bundles (Goodlett et al., 2009; Colby et al., 2012), and popularized by automated pipelines such as AFQ (Yeatman et al., 2012) and the UNC-Utah NA-MIC framework (Verde et al., 2014), with tract segmentation by learned models in recent pipelines (Wasserthal et al., 2018). Tractometry has been extended to many measures at once (Chamberland et al., 2019), and to multivariate and machine-learning analyses of all profiles of a scan (Richie-Halford et al., 2021). Its reliability has been studied (Kruper et al., 2021), but to our knowledge the imputation of missing profile values has not been systematically evaluated.

Missing-data imputation. Classical results distinguish missing completely at random, at random and not at random (Rubin, 1976; Little and Rubin, 2019). Practical imputation methods include k-nearest neighbours (Troyanskaya et al., 2001), multiple imputation by chained equations (van Buuren and Groothuis-Oudshoorn, 2011), random forests (Stekhoven and Bühlmann, 2012) and low-rank models; probabilistic principal component analysis (PPCA) handles missing values in a principled way through the EM algorithm (Tipping and Bishop, 1999; Ilin and Raiko, 2010). Deep generative models amortize inference over incomplete data: the partial VAE of EDDI (Ma et al., 2019) and MIWAE (Mattei and Frellsen, 2019). Masked autoencoders (He et al., 2022) learn to reconstruct hidden parts of an input with transformers (Vaswani et al., 2017) and are, in effect, amortized imputers. For a broader review in pattern recognition see García-Laencina et al. (2010).

Normative models. Normative modeling characterizes the distribution of a brain measure in a reference population as a function of covariates such as age, and expresses an individual as deviations from it (Marquand et al., 2016; Rutherford et al., 2022; Bethlehem et al., 2022). Our phase 2 imputes such deviations instead of raw values; phase 3 adds a joint low-rank model of the deviations across all locations and measures.

Neural fields and functa. Sine-activated networks (SIREN) represent signals as continuous functions of coordinates (Sitzmann et al., 2020). Functa (Dupont et al., 2022) and its medical variant MedFuncta (Friedrich et al., 2025) share one network across a dataset and fit a per-instance latent modulation, so that the shared network acts as a prior; we test both a per-scan SIREN and a functa model.

Graph networks. Graph convolutional networks (Kipf and Welling, 2017) and graph U-Nets (Gao and Ji, 2019) operate on irregular structures such as the locations of fiber tracts. CONFETTI (Styner, 2026) builds a multi-resolution neighbourhood graph over tract profile locations for scan classification; we adapt it to imputation.

Harmonization. Multi-site pooling of diffusion measures requires correcting site or acquisition effects, classically by location-and-scale models such as ComBat (Johnson et al., 2007; Fortin et al., 2017); our batch correction is an age-aware location-and-scale model of this kind.

3 Materials and methods

3.1 Data

Cohorts. The normative cohort is the BCP (Howell et al., 2019): 423 scans of 208 children, aged 0 to 65 months (median 13, quartiles 8 and 21; Table 1, Fig. 1a). 105 children contributed two or more scans (up to 9). Scans fall in three age bins used throughout: 0–3 months (28 scans), 4–9 months (117) and 10–60 months (278). The scan names carry three acquisition labels (`dir79select`, 311 scans; `shells06select`, 108; `shells06`, 4). [TODO: BCP acquisition details (scanner, b-values, directions, resolution), preprocessing pipeline, tensor and NODDI fitting, ethics approval and consent statements.] The second cohort is the HBCD study (Volkow et al., 2021; Dean et al., 2024): 2,293 scans of 1,720 children, median age 1.4 months (range -1.5 to 15.8; 1,501, 658 and 133 scans in the three bins, and 1 outside them). [TODO: HBCD release, age definition (the negative ages suggest age relative to a reference date such as term-equivalent age), acquisition, processing and data use agreement.]

Table 1: The two cohorts (computed from the profile tables).

	BCP	HBCD
Scans	423	2,293
Subjects	208	1,720
Subjects with ≥ 2 scans	105	512
Scans per subject, median (max)	2 (9)	1 (3)
Age, months: median [IQR]	13.0 [8.0, 21.0]	1.4 [0.9, 5.8]
Age range, months	0.0–65.0	-1.5–15.8
Scans aged 0–3 / 4–9 / 10–60 mo	28 / 117 / 278	1,501 / 658 / 133
Scans outside the age bins	0	1
Tracts / profile locations	69 / 5,388	69 / 5,388
Measures	7	7

Profiles. Profiles were extracted on the 69 tracts of the DTI_IBISEP_Feb26 atlas [TODO: atlas reference], sampled every 1 mm of arc length: 5,388 profile locations per measure. Each location has a 3-D position in atlas space, the point of the tract's axis (computed from the atlas fibers with `dmrifiberprofile compute-axis`) at that arc length. Tract axes are dense: the median distance from a tract's locations to the nearest other tract's axis is 2.4 mm for the median tract, and 37% of all locations are within 2 mm (76% within 4 mm) of another tract. Seven measures were analysed: FA, MD, radial and axial diffusivity (RD, AD) of the diffusion tensor, and NDI, ODI and FWF of NODDI (Zhang et al., 2012) (FWF is the isotropic fraction of the NODDI fit). The free-water tensor measures (Pasternak et al., 2009), imputed in the early phases, were left out after the free-water fits failed in many HBCD scans (Section 4.7). The profiles were cleaned by the QC of DTIPlayground (`qc-profiles`: outlier profiles removed, diffusivities above 3.6×10^{-3} mm^2/s removed as single values) [TODO: describe the QC criteria in more detail].

Missingness of the real profiles. After QC, 2.29% of the BCP cells (365,899 of 15,953,868 scan × location × measure values) are missing. Nearly all (99.8%) belong to whole missing profiles: 4,599 of 204,309 scan × tract × measure profiles (2.25%), always all measures of a tract at once (657 rejected scan × tract combinations). Only 168 profiles are partly missing: 142 miss their start or end (median 3, at most 9 locations) and 26 have an interior gap. 203 of the 423 scans miss at least one profile value; 2 scans miss more than half of their tracts and are excluded by default. The share of scans with a given tract rejected ranges up to 6.1% (AC_olfactory; median over tracts 2.1%; Fig. 1b). No BCP scan lacks a whole measure family after QC, but this pattern is the norm in cohorts where one model fit fails or is not computed.

3.2 Notation and the age-bin norm

For scan s, location p (tract, arc length) and measure m, y_{spm} is the profile value. The age-bin norm is the mean μ_{bpm} and SD σ_{bpm} of the cohort's scans of age bin b; the deviation of a scan is $z_{spm} = (y_{spm} - \mu_{b(s)pm})/\sigma_{b(s)pm}$. Methods "on the deviations" impute z and return $\hat{y} = \mu + \sigma\hat{z}$ with the norm of the missing location. In validation, the norm is always computed without every scan of the validated scan's subject.

3.3 Imputation methods

Table 2 summarizes the methods, grouped by the six phases of the evaluation. All are implemented in `dtiplayground/dmri/fiberprofile/analysis/impute_profiles.py` (with `graph_unet.py`, `amortized.py` and `functa.py`).

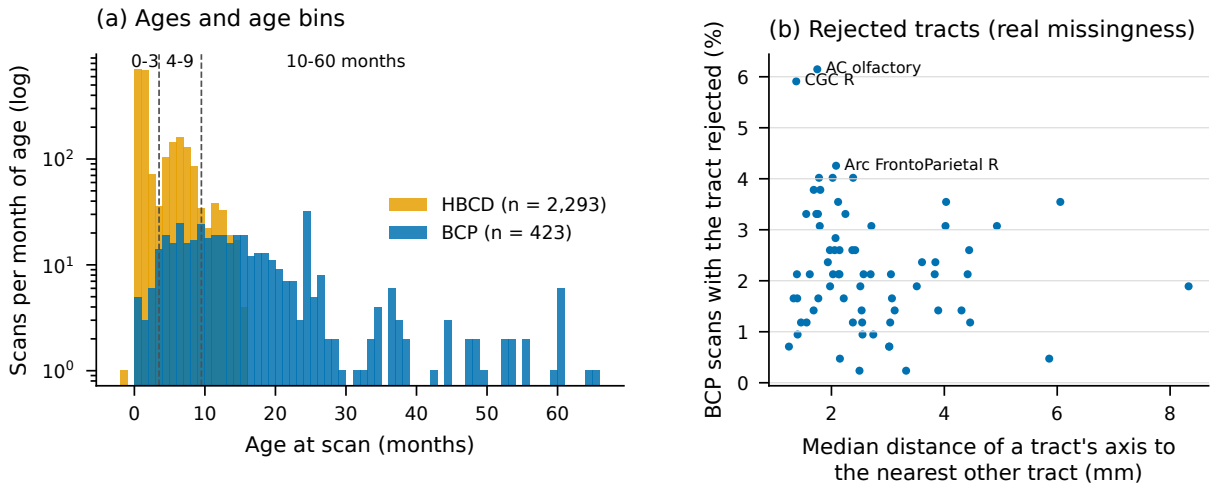


Figure 1: Data. (a) Ages at scan of BCP (normative cohort) and HBCD (reference cohort); dashed lines: the age bins. (b) Real missingness of BCP after QC: share of scans with a tract rejected (all measures missing), against the median distance from the tract's axis to the nearest other tract.

Phase 1: the scan's own values. *Arc-linear* interpolates linearly along the arc length between the observed values at both ends of a gap (gaps up to 15 mm); missing ends are left, filled with the nearest observed value (*constant*) or extrapolated from the three nearest. *Linear* interpolates barycentrically within the Delaunay tetrahedra of the observed 3-D positions of the same measure in all tracts. *kNN* averages the k nearest observed positions with inverse-distance weights. *SIREN* (Sitzmann et al., 2020) fits, per scan, a sine network (3 hidden layers of 256 units, Adam, learning rate 10^{-4}; Kingma and Ba, 2015) to all observed values of all measures jointly, each standardized within the scan; its inputs are the 3-D position scaled to $[-1, 1]$ (*xyz*), optionally with the arc length (*xyz+arc*) or a one-hot tract code (*xyz+tract*); we varied $\omega_0 \in \{5, 10, 20, 30\}$ and 100–1000 epochs.

Phase 2: deviations from the age-bin norm. The same methods applied to z instead of y. Values farther than a distance limit from any observed value of the same measure (20 mm on the deviations) are left to the age-bin mean.

Phase 3: population model. The deviations of the normative scans, over all locations and measures jointly (5,388 × 7 cells observed in at least 10 scans), are modelled by PPCA (Tipping and Bishop, 1999), $\mathbf{z} = \boldsymbol{\mu} + W\mathbf{x} + \boldsymbol{\epsilon}$, $\mathbf{x} \sim \mathcal{N}(0, I_q)$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \psi I)$, fitted by EM on the observed cells with the exact posterior of every row (Ilin and Raiko, 2010), a ridge prior on the rows of W, and $q = 80$ components by default (5–160 tested; isotropic, per-measure and per-cell noise tested). A scan's observed deviations give the posterior mean and SD of every missing cell, including all cells of a measure family the scan lacks. The posterior SDs are rescaled per measure so that errors on held-out whole profiles (5 folds by subject) have unit root-mean-square z. The *PPCA + local* combination uses the PPCA prediction as the scan's own norm and imputes the residuals with SIREN (xyz, $\omega_0 = 10$, 500 epochs) or kNN ($k = 20$) within 4 mm of an observed value; beyond, the PPCA prediction is used. This combination (PPCA + SIREN) was the default of DTIPlayground up to version 0.9.5.

Phase 3b: continuous age norms and pooling. Instead of age bins, the mean and SD of every cell can be smooth functions of log(age + 1 month): a natural cubic spline or a local-linear kernel regression with leave-one-out bandwidth. To pool cohorts, each batch is mapped to the

Table 2: Methods and what they use. Patterns: G interior gap, E end gap, W whole profile, R region, F whole measure family. "Own": values of the scan itself; "norm": age-bin norm; "cohort": models fitted on the normative cohort (without the validated subject).

Method	Information	Description	Patterns
Arc-linear	own	linear interpolation along the arc length of the same profile; ends left, repeated (constant) or extrapolated	G, E
Linear (3-D)	own	barycentric interpolation in the Delaunay tetrahedra of the scan's observed locations of the same measure in all tracts	G, E, W, R
kNN	own	inverse-distance-weighted mean of the k nearest observed locations in 3-D	G, E, W, R
SIREN	own	per-scan sine network, inputs 3-D position (xyz), plus arc length or one-hot tract; all measures jointly	G, E, W, R
Phase 2: any of the above on z	own + norm	imputes deviations from the age-bin norm	as above
Population model (PPCA)	own + cohort	probabilistic PCA of the deviations of all locations and measures jointly; posterior mean and SD of the missing cells	all
PPCA + local	own + cohort	SIREN or kNN imputes the residuals from the PPCA prediction within 4 mm of an observed value; PPCA beyond	all
Graph U-Net (hybrid)	own + cohort	multi-resolution graph convolutional U-Net over the locations; corrects the PPCA prediction	all
Transformer (hybrid)	own + cohort	masked autoencoder over patches of 8 locations; corrects the PPCA prediction	all
Blend	own + cohort	$(1-w)\,\hat{y}_{\text{PPCA+SIREN}} + w\,\hat{y}_{\text{model}}$, w fixed, by precision, or per measure and distance band	all
Explored, not adopted			
Functa	own + cohort	shared SIREN, per-scan latent fitted by meta-learned gradient steps	all
Partial VAE	own + cohort	set encoder of the observed cells, per-location decoder (EDDI/MIWAE-style)	all
Age-dependent PPCA	own + cohort	PPCA per age, scans weighted by a kernel of log age	all
Acquisition correction	own + cohort	random-intercept offset per acquisition label, removed before imputation	all

reference batch by an age-aware shift and scale per cell and measure (one spline age curve for all batches plus a shift per batch, fitted jointly; the scale as the ratio of residual SDs over the common age range), and imputed values are mapped back.

Phase 4: graph U-Net. After CONFETTI (Styner, 2026), a multi-resolution graph keeps every 2^L-th location of each tract at level L (4 levels); nodes are linked along the tract and to the closest node of each other tract within 1.5 times the median distance to the nearest other tract's axis, with Gaussian edge weights. A U-Net (Gao and Ji, 2019) of graph convolutions (Kipf and Welling, 2017) (128 features, one convolution per level, batch normalization, dropout 0.1, skip connections) receives per location the observed deviations (0 where missing), the observation mask, the 3-D position and arc length, and, in the *hybrid* variant, the PPCA prediction and its log SD; a scan-wide summary (mean and max at the coarsest level) is added there. It outputs a mean and log variance per location and measure and is trained (AdamW, Loshchilov and Hutter, 2019; cosine schedule; batches of 16 scans; up to 60 epochs) with the validation patterns of Section 3.4 hidden

anew in every epoch (a region in half of the scans, a whole family in a quarter), on the Gaussian negative log-likelihood of the hidden values (Kendall and Gal, 2017) (visible values with weight 0.1). Ten percent of the training subjects select the epoch and calibrate the SDs. The PPCA input of a training scan comes from a model fitted without its subject (5-fold cross-fitting). We also evaluated CONFETTI's original configuration (graph threshold from the dense atlas fibers, 64 features, no scan-wide summary).

Phase 6: transformer. A masked-autoencoder transformer (Vaswani et al., 2017; He et al., 2022): each tract is cut into patches of 8 consecutive locations (about 700 tokens per scan); a token embeds the per-location inputs of its patch (as for the U-Net) plus a learned tract embedding; 6 pre-norm self-attention layers of width 192 (6 heads, dropout 0.1) relate every token to every other, and a linear head returns a mean and log variance for every location of the patch. Training is as for the graph U-Net (AdamW, learning rate 10^{-3}, weight decay 10^{-4}, cosine schedule, 200 or 400 epochs with the best held-out epoch kept), in the hybrid variant correcting a cross-fitted PPCA prediction.

Phase 6: blends. The PPCA + SIREN prediction b and a model prediction g (U-Net or transformer) are combined as $(1-w)b + wg$, with (i) fixed weights w_{near} for cells within 4 mm (those SIREN fills) and w_{beyond} for the others; (ii) precision weights $w = \sigma_b^{-2}/(\sigma_b^{-2} + \sigma_g^{-2})$; or (iii) a *weight table* per measure m and band k of the distance to the nearest observed value of the measure (0–0.5, 0.5–1, 1–2, 2–3, 3–4 mm for SIREN cells; 4–6, 6–10, 10–20, >20 mm for fallback cells; and a measure the scan has no observed value of). The table is the weighted least-squares solution in deviation units,

$$w_{mk} = \text{clip}_{[0,1]} \frac{\sum_{i \in (m,k)} c_i \, (t_i - b_i)(g_i - b_i)}{\sum_{i \in (m,k)} c_i \, (g_i - b_i)^2}, \tag{1}$$

over hidden cells i with true deviation t_i, where c_i is the inverse mean squared error of the age-bin mean in the cell's (pattern, measure, age bin) group, so that each group counts as in the score. This is a constrained form of stacking (Wolpert, 1992; Breiman, 1996). The table shipped with DTIPlayground 0.9.8 was fitted on all validation cells of the three draws; all scores reported for it here are cross-fitted (weights for a scan fitted without its subject, 5 folds). We also tested a gradient-boosted weight per cell and calibrated precision weights.

Reference cohort. With `-transformer-reference-dir`, a second cohort (HBCD) is batch-corrected to BCP at the same age and added to the training of the transformer only; the norm, PPCA and SIREN remain those of BCP. We compared this with pooling BCP and HBCD for the norm, the PPCA and the transformer.

Explored alternatives. *Functa* (Dupont et al., 2022; Friedrich et al., 2025): one SIREN shared by the normative scans, modulated per scan by shifts from a latent vector fitted by a few meta-learned gradient steps on the scan's observed values, meta-trained on the hidden values of the validation patterns; we varied inner steps and step sizes, test-time refinement, an encoder initialization, depth (2–8), width (64–512), latent size (16–512), age and tract inputs, and graph variants (Laplacian-eigenvector inputs, a learned vector per location, a graph-convolution decoder), each on the deviations and as a hybrid correcting PPCA. *Partial VAE* (Ma et al., 2019; Mattei and Frellsen, 2019): a set encoder of the observed cells (per-location MLP pooled by mean and max), a latent (32–128), a decoder with weights per location, importance-weighted training. *Age-dependent PPCA*: a PPCA per age, the scans weighted by a Gaussian kernel of log age. *Acquisition correction*:

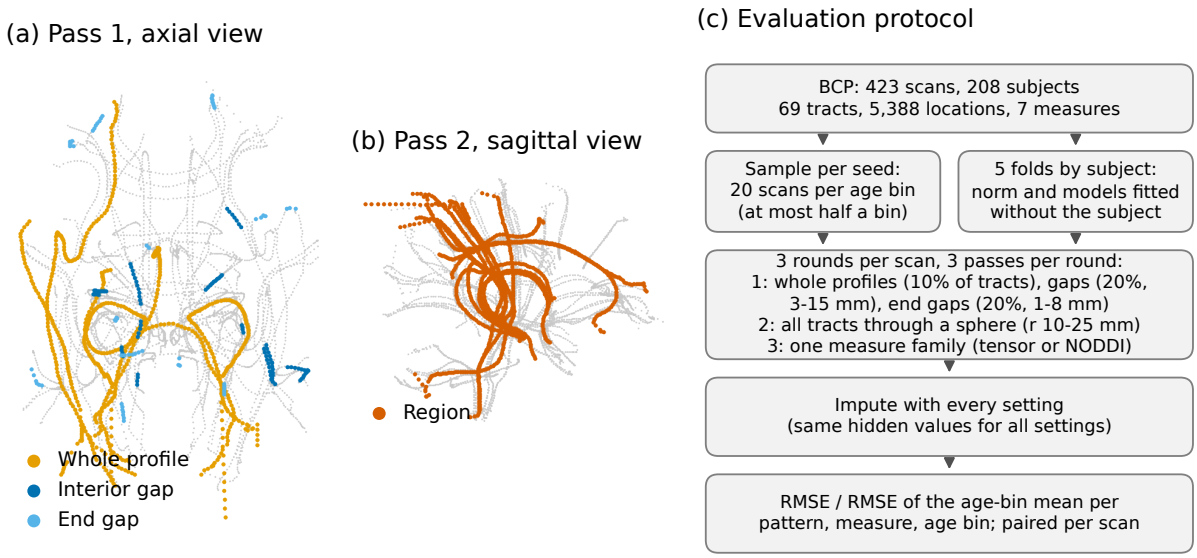


Figure 2: Missingness patterns and evaluation protocol. (a, b) One round of hidden values on a 12-month BCP scan, the profile locations projected on axial and sagittal planes (grey: observed). (a) Pass 1 hides whole profiles of random tracts, interior gaps and end gaps; (b) pass 2 hides all tracts through a sphere (this draw: 13 of 69 tracts). Pass 3 (not shown) hides a whole measure family. (c) The protocol.

an offset per acquisition label and cell fitted on the deviations with a random intercept per subject, removed before imputation. *Context inputs* to the transformer: distance to the nearest observed value, age and acquisition. *Seed ensembles*: three transformers averaged (Lakshminarayanan et al., 2017).

3.4 Evaluation protocol

Hiding patterns. Measured values are hidden, imputed, and compared with the truth (Fig. 2). Each validated scan goes through three rounds; each round has three passes on the complete scan: (1) whole profiles (all measures) of 10% of the tracts with at least three observed locations; of the remaining tracts, an interior gap of 3–15 mm in 20% (*gap*) and the first or last 1–8 mm of the observed part in another 20% (*end gap*; at least three observed locations kept); (2) every tract with an observed location inside a sphere of radius 10–25 mm around a random observed location loses its whole profile (*region*; spheres reaching more than half of the tracts are redrawn), mimicking QC rejection after a regional artifact; (3) all values of one measure family the scan has, the tensor family (FA, MD, RD, AD) or the NODDI family (NDI, ODI, FWF), chosen at random (*whole family*). The hiding is identical for all settings of a run and across runs with the same random seed; whole profiles and regions mirror the dominant real missingness (Section 3.1), end gaps its most common partial pattern.

Sampling and folds. Per random seed (0, 1, 2), 20 scans per age bin were drawn, at most half of a bin (14, 20 and 20: 54 scans), 131 distinct scans of 96 subjects over the three seeds, 162 scan × seed pairs. Every model fitted on the cohort (norm, PPCA, continuous norms, batch correction, networks, blend weights) is fitted without the validated subjects: the validated scans are split by subject into 5 folds and each fold's models are fitted on all other subjects.

Score. For a setting, pattern P, measure m and age bin b, the RMSE over the hidden values is divided by the RMSE of the age-bin mean on the same values,

$$\rho_{Pmb} = \mathrm{RMSE}_{Pmb}(\text{setting})/\mathrm{RMSE}_{Pmb}(\text{age-bin mean}), \tag{2}$$

so that $\rho < 1$ means better than the norm. Values a setting leaves unfilled (e.g. whole profiles for arc-linear, cells beyond a distance limit, a whole family for same-scan methods) are counted with the age-bin mean, as an imputation with that fallback would do. The pattern score is the mean of ρ over measures and age bins; the *overall* score is the mean over the four patterns gap, end gap, whole profile and region; the whole-family pattern is reported separately for the tensor and the NODDI family. Scores are averaged over the three seeds (SD between seeds in parentheses). Absolute RMSEs are reported in the units of each measure.

Paired statistics. Seed-to-seed variation of a score (SD up to about 0.01 overall and more within an age bin) is as large as many differences between settings, so settings are compared on the same scans. The per-scan score is the mean over pattern × measure of $\sqrt{\mathrm{MSE}/\mathrm{MSE}_{\text{age-bin mean}}}$ for that scan; for two settings A (reference) and B, we report the geometric mean over scan × seed pairs of the ratio of their per-scan scores, its 95% confidence interval from the normal approximation of the log ratios, the Wilcoxon signed-rank test (Wilcoxon, 1945) of the log ratios, and the share of pairs in which B is better. The 162 pairs are not independent: the same scan can be drawn by several seeds, and several scans come from one subject. We therefore also report a cluster bootstrap CI that resamples the 96 subjects (4,000 resamples; Field and Welsh, 2007) and a Wilcoxon test on per-subject mean log ratios, and we Holm-adjust (Holm, 1979) the p values over the 19 pre-specified contrasts of Table 7. Paired tests are only reported between settings that fill every hidden value.

Quick screens. Model families that failed to justify a full validation (functa, partial VAE, age-dependent PPCA) and transformer variants were screened on one fixed held-out split: 30 test subjects, two rounds of the same hiding patterns, models fitted on the other subjects, and the same score on the 7 measures. Screen numbers are comparable within the screen but not directly with the 3-seed validation.

Earlier protocols. Phases 1–3 were first run on 30 scans, one seed and 11 measures (including the free-water tensor measures), and phase 3b on the 11 measures with three seeds. All phases 1–4 were later re-run with the protocol above; we report those results throughout, and use the earlier runs only where no re-run exists (the SIREN hyperparameter grid, the distance limit of the PPCA residuals and the pooling experiment; Appendix).

3.5 Implementation and reproducibility

All methods are part of DTIPlayground 0.9.8 (NIRAL, 2026) (PyTorch, Paszke et al., 2019). Each validation run is a single `dmrifiberprofile impute-profiles -validate` call, with `-validate-seed` selecting the sample and the hidden values; its outputs include per-scan, per-age-bin and per-distance error tables and, with `-validate-dump-dir`, every hidden cell with all predictions, from which the blends were fitted offline. A run with the population model, SIREN and one network took about 2 hours on one NVIDIA RTX A6000 GPU. Every number, table and figure of this paper is produced from those result files by the scripts accompanying the paper (Data and code availability).

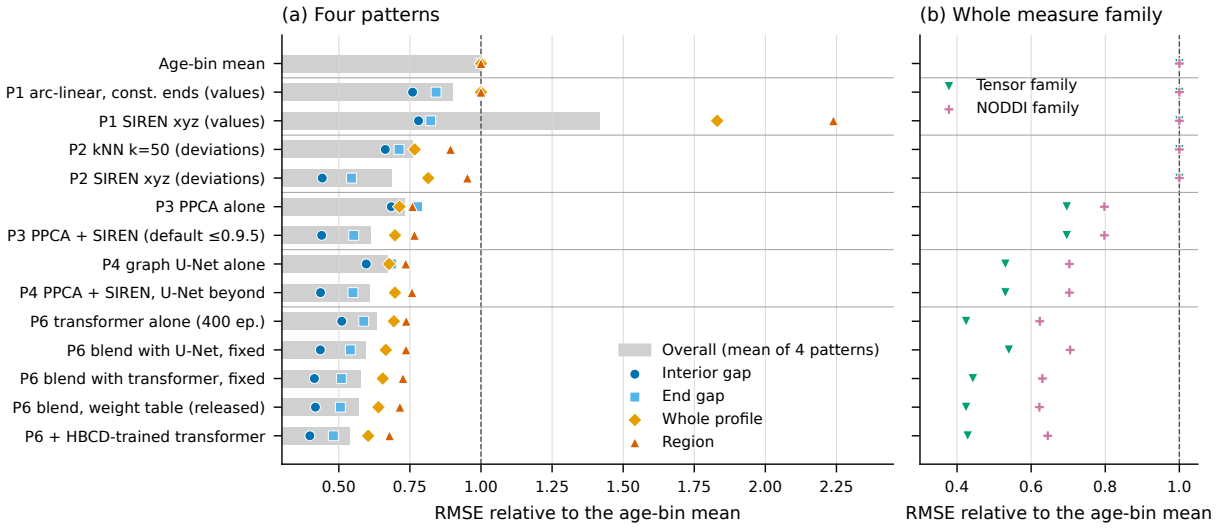


Figure 3: Phase-wise progression. (a) Error relative to the age-bin mean per pattern (markers) and overall (bars). (b) A whole measure family hidden; methods of phases 1–2 cannot impute it (1.0 by construction). P1–P6: phases; "released": the default of DTIPlayground 0.9.8 (scored cross-fitted).

4 Results

4.1 Overview: from the scan's own values to blended models

Table 3 and Fig. 3 give the error of representative settings of every phase on the same hidden values.

Phase 1: the scan alone is not enough. Imputed from the scan's own values, only interior gaps were better than the age-bin mean (arc-linear 0.759, SIREN xyz 0.780); with constant ends, arc-linear also improved end gaps (0.842). Whole profiles and regions were imputed far worse than the norm by every 3-D method (SIREN xyz 1.831 and 2.239, linear 1.818 and 2.123). Neighbouring tracts differ systematically in their measures, by orientation and organization, so a missing tract cannot be read off its neighbours' values. The best phase-1 setting overall, arc-linear with constant ends (0.900), owes this to filling nothing beyond gaps and ends.

Phase 2: the norm makes every pattern imputable. On the deviations from the age-bin norm, every 3-D method was better than the age-bin mean for whole profiles, and all but kNN with $k = 1$ for regions. The best single setting was SIREN with the 3-D position only (0.688 overall; gaps 0.442, end gaps 0.544, whole profiles 0.814, regions 0.952). A scan that deviates from the norm in one tract tends to deviate the same way in its neighbours, while the norm carries the tract-specific level that the neighbours cannot. The gain shrank as more was missing: regions remained close to the norm.

Phase 3: a population model. PPCA alone (0.733) was better than phase-2 SIREN for whole profiles (0.713 vs 0.814) and regions (0.759 vs 0.952), but worse for gaps and ends, where neighbours along the same profile carry more. Imputing the PPCA residuals locally combined both: PPCA + SIREN (0.614; gaps 0.440, end gaps 0.552, whole profiles 0.697, regions 0.766) was better than PPCA alone on every scan draw (paired ratio 0.834, Table 7) and better than PPCA + kNN (0.664; kNN was 1.082 times SIREN's error, although slightly better for whole profiles and regions). PPCA is also the first method that imputes a missing measure family (tensor 0.696, NODDI 0.798).

Table 3: Error relative to the age-bin mean (RMSE ratio ρ; below 1: better) on identical hidden values, mean over 3 seeds × 54 BCP scans, 7 measures and 3 age bins. Overall: mean of the four patterns (SD between seeds). †Leaves some hidden values, counted with the age-bin mean (a whole family is never filled by phases 1–2). *Weights cross-fitted by subject; the mean of the per-seed scores (pooled over seeds, as in the software documentation: 0.567 and 0.538).

		Pattern				Whole family missing	
Setting	Overall (SD)	Gap	End	Whole	Region	Tensor	NODDI
Reference							
Age-bin mean	1.000 (0.000)	1.000	1.000	1.000	1.000	1.000	1.000
Phase 1: the scan's own values							
Arc-linear, constant ends†	0.900 (0.015)	0.759	0.842	1.000	1.000	1.000	1.000
Linear (3-D Delaunay)†	1.496 (0.043)	1.074	0.968	1.818	2.123	1.000	1.000
kNN, $k = 20$†	1.713 (0.048)	1.490	1.198	1.893	2.269	1.000	1.000
SIREN, xyz†	1.418 (0.045)	0.780	0.823	1.831	2.239	1.000	1.000
SIREN, xyz+tract†	1.572 (0.057)	0.732	0.812	2.335	2.408	1.000	1.000
Phase 2: deviations from the age-bin norm							
Arc-linear, constant ends†	0.750 (0.011)	0.431	0.570	1.000	1.000	1.000	1.000
Linear (3-D Delaunay)†	0.733 (0.006)	0.496	0.701	0.801	0.934	1.000	1.000
kNN, $k = 20$†	0.755 (0.009)	0.631	0.683	0.782	0.923	1.000	1.000
kNN, $k = 50$†	0.759 (0.010)	0.663	0.712	0.767	0.893	1.000	1.000
SIREN, xyz+arc†	0.704 (0.005)	0.422	0.516	0.895	0.982	1.000	1.000
SIREN, xyz†	0.688 (0.004)	0.442	0.544	0.814	0.952	1.000	1.000
Phase 3: population model (PPCA, $q = 80$)							
PPCA alone	0.733 (0.008)	0.685	0.777	0.713	0.759	0.696	0.798
PPCA + kNN on residuals (≤4 mm)	0.664 (0.008)	0.568	0.659	0.675	0.752	0.696	0.798
PPCA + SIREN on residuals (≤4 mm)	0.614 (0.007)	0.440	0.552	0.697	0.766	0.696	0.798
Phase 4: graph U-Net (hybrid)							
U-Net alone	0.673 (0.008)	0.596	0.686	0.676	0.735	0.530	0.703
U-Net + SIREN on residuals	0.659 (0.014)	0.485	0.629	0.741	0.780	0.530	0.703
PPCA + SIREN, U-Net beyond 4 mm	0.610 (0.007)	0.436	0.550	0.697	0.757	0.530	0.703
Phase 6: transformer and ensembles							
Transformer alone, 200 epochs	0.654 (0.009)	0.547	0.627	0.704	0.738	0.446	0.636
Transformer alone, 400 epochs	0.632 (0.005)	0.510	0.588	0.694	0.737	0.425	0.623
PPCA + SIREN, transformer beyond 4 mm	0.606 (0.008)	0.438	0.542	0.692	0.750	0.446	0.636
Blend with U-Net, fixed 0.25/0.75	0.594 (0.006)	0.435	0.541	0.665	0.736	0.540	0.705
Blend with transformer (200), fixed 0.25/0.75	0.582 (0.008)	0.424	0.518	0.658	0.729	0.460	0.643
Blend with transformer (400), fixed 0.25/0.75	0.576 (0.006)	0.414	0.509	0.654	0.726	0.443	0.630
Blend with transformer (400), weight table*	0.569 (0.005)	0.418	0.505	0.639	0.715	0.424	0.622
Phase 6: transformer (200 epochs) trained with the HBCD reference cohort							
Reference transformer alone	0.580 (0.013)	0.456	0.532	0.633	0.698	0.432	0.662
Blend, fixed 0.25/0.75	0.557 (0.009)	0.400	0.495	0.634	0.697	0.439	0.642
Blend, weight table*	0.540 (0.010)	0.398	0.481	0.603	0.678	0.429	0.645

Phase 4: graph U-Net. In CONFETTI's original configuration the graph U-Net imputed poorly (0.90 of the age-bin mean's error overall in a development run on one seed and 11 measures, against 0.69 for PPCA alone; software documentation). Three changes made it competitive: a graph built from the tract axes rather than the dense fibers (fewer edges to other tracts), a scan-wide context at the coarsest level, and correcting the PPCA prediction (hybrid). The hybrid U-Net alone (0.673) was better than PPCA alone for every pattern but worse than PPCA + SIREN for gaps and ends (0.596 vs 0.440); adding SIREN on its residuals did not close the gap (0.659). Its strength was a missing family: 0.530 for the tensor and 0.703 for the NODDI family, against 0.696 and 0.798 for PPCA. Used only beyond 4 mm (PPCA + SIREN near, U-Net beyond) it changed the four patterns little (0.610; paired 0.993).

Phase 6: transformer and blends. The transformer alone (0.654 with 200 epochs, 0.632 with 400) was better than the U-Net alone (paired 0.976) and better still for a missing family (tensor 0.425, NODDI 0.623 with 400 epochs). Neither network alone reached PPCA + SIREN on the four patterns (transformer 1.059, U-Net 1.085 times its error), because SIREN on the PPCA residuals is better close to observed values. Blending did: a fixed blend with weight 0.25 near and 0.75 beyond 4 mm reduced the error to 0.594 with the U-Net and 0.582 with the transformer (0.576 with 400 epochs), and the learned weight table to 0.569 (cross-fitted). Against PPCA + SIREN, the released default (transformer of 400 epochs, weight table) had a paired ratio of 0.923 and was better on 99% of scan draws; it improved every pattern (gaps 0.418, end gaps 0.505, whole profiles 0.639, regions 0.715) and a missing family by far (tensor 0.424, NODDI 0.622).

4.2 Which approach works for which pattern

Interior gaps and end gaps. Gaps are the easiest pattern: 93% of the hidden values are within 4 mm of an observed value (85% for end gaps, 72% for whole profiles, 32% for regions; Fig. 4d), and local methods on deviations or PPCA residuals win. Without a norm, arc-linear interpolation along the profile is a reasonable choice for interior gaps (0.759); with a norm, arc-linear on the deviations reached 0.431 and SIREN on the PPCA residuals 0.440. End gaps behave like gaps with one-sided support and were consistently about 0.1 worse. Amortized networks alone were clearly worse here (transformer 0.510), but contributed in the blend (0.418).

Whole profiles and regions. A whole profile is imputed from neighbouring tracts and from the scan's other tracts and measures; a region also removes the neighbours. Here the population model and the amortized networks mattered: PPCA alone 0.713 and 0.759, phase-2 SIREN 0.814 and 0.952, the transformer alone 0.694 and 0.737, the released blend 0.639 and 0.715. Regions remain the hardest of the four patterns for every method.

A whole measure family. Only cohort models can impute a measure the scan lacks entirely, from the scan's other measures. The amortized networks were much better than PPCA (tensor family: transformer 0.425, U-Net 0.530, PPCA 0.696; NODDI family: 0.623, 0.703, 0.798); paired over scans, the transformer's error for a missing family was 0.667 of PPCA's (400 epochs, better on 100% of scan draws). The tensor family was easier to recover from NODDI than the reverse, and MD and RD were recovered best (per measure: Table 5).

Distance to the nearest observed value. Within each pattern, the error grew with the distance to the nearest observed value of the same measure (Fig. 4). The deviations of a far neighbour still carried information on the deviations scale (phase-2 SIREN stayed below 1 up to its 20 mm limit for regions), but the PPCA residuals are local: SIREN on them helped within a

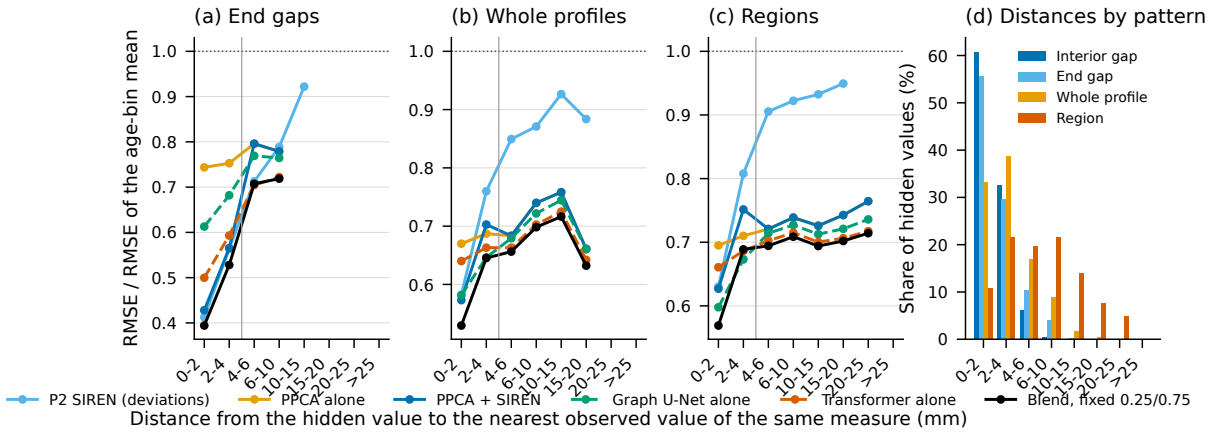


Figure 4: Error by the distance from a hidden value to the nearest observed value of the same measure, per pattern (3 seeds; bins with fewer than 600 hidden values omitted; vertical line: 4 mm). (d) The distribution of these distances per pattern. Interior gaps are omitted from (a)–(c): 93% of their values lie within 4 mm.

few millimetres and hurt beyond, which is why it is limited to 4 mm (Appendix Fig. A2). Beyond 4 mm the amortized networks were better than PPCA, which is what the blend exploits (Fig. 5c).

4.3 How much does the normative model help?

The normative information enters at three levels. The age-bin norm alone turned SIREN's whole profiles from 1.831 (phase 1, on values) into 0.814 (phase 2, on deviations), its regions from 2.239 into 0.952, and its overall score from 1.418 to 0.688. The population model, which captures how deviations co-vary across all tracts and measures, took the overall error from 0.688 to 0.614 and made a missing family imputable. Learned amortized models of the deviations took it to 0.569 (or 0.540 with a larger training cohort). In absolute terms (Table 6), the released default roughly halved the RMSE of the age-bin mean for the diffusivities.

A continuous age norm (spline or kernel regression of log age) was a better predictor than the age-bin mean on its own (paired 0.930, 95% CI 0.911–0.949; score 0.868 in 0–3 months, where the norm changes fastest within a bin; Appendix Table A1), but under PPCA + SIREN it made no difference (1.005, CI 0.996–1.015): the population model and the scan's own values absorb the within-bin age trend.

4.4 Population model components and local methods

The PPCA error fell up to 80 components and was flat beyond (overall alone: 0.789 with 10, 0.752 with 40, 0.733 with 80, 0.732 with 120). Combined with SIREN, 40 and 120 components were slightly but consistently worse than 80 (paired 1.008 and 1.010). The three noise models differed by less than 0.01 in the earlier 30-scan runs (software documentation), so the isotropic model was kept. Within SIREN, the trade-off between patterns was clear in the hyperparameter grid (Appendix Fig. A1): gaps and ends need long training and a high ω_0 or a tract code, whereas whole profiles and regions are best after 100 epochs and get worse as the network fits the observed tracts more closely; xyz with $\omega_0 = 10$ and 500 epochs balanced them best on the deviations.

4.5 Blend weights

The learned weight of the transformer rose with the distance to the nearest observed value (Fig. 5): 0.16–0.27 next to an observed value (where SIREN on the PPCA residuals dominates), about 0.6 from 3–4 mm, 0.57–0.73 for fallback cells, and 0.82–0.98 for a missing family (BCP-trained

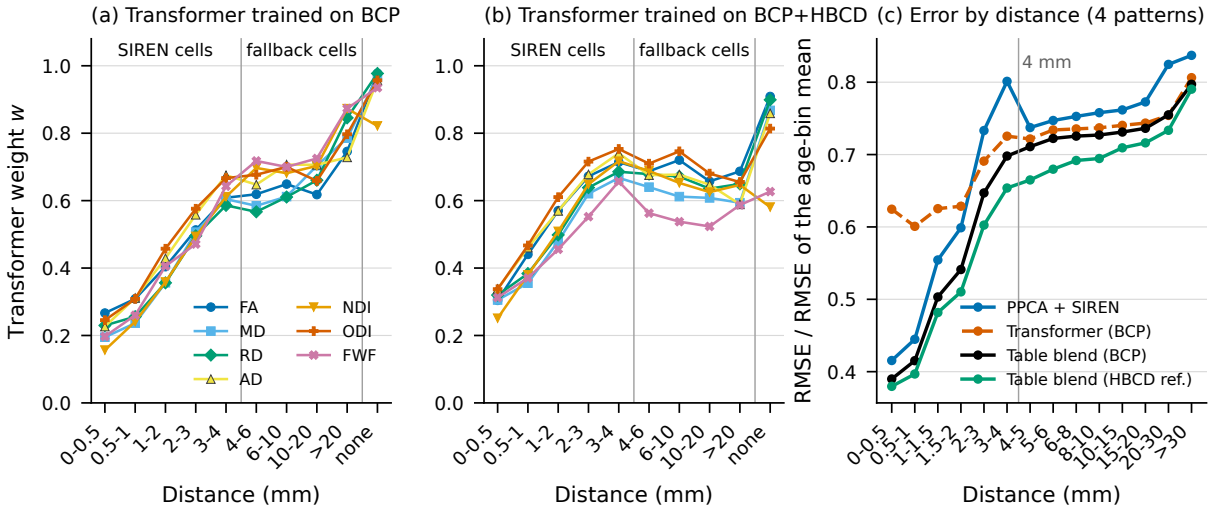


Figure 5: Blend weights. (a, b) Weight w of the transformer per measure and distance band (SIREN cells within 4 mm, fallback cells beyond, "none": the scan has no observed value of the measure), fitted on all validation cells; (a) transformer trained on BCP (the default table), (b) trained on BCP and HBCD. (c) Error relative to the age-bin mean against distance, four patterns pooled, for PPCA + SIREN, the BCP transformer alone and the cross-fitted table blends.

Table 4: Quick held-out screen (one split, 30 test subjects, 2 rounds; 7 measures): best result per model family and target; "Runs": number of configurations and test-time variants screened. Not directly comparable with Table 3.

Model	Target	Runs	Overall	Gap	End	Whole	Region	Tensor	NODDI
PPCA ($q = 80$)	–	1	0.720	0.665	0.780	0.702	0.732	0.683	0.774
Age-dependent PPCA	–	3	0.732	0.685	0.792	0.712	0.737	0.704	0.783
Partial VAE	deviations	4	0.807	0.783	0.844	0.798	0.801	0.803	0.885
Partial VAE	PPCA correction	4	0.730	0.673	0.789	0.712	0.745	0.696	0.786
Functa	deviations	93	0.736	0.674	0.752	0.730	0.787	0.692	0.836
Functa	PPCA correction	94	0.702	0.641	0.750	0.689	0.728	0.671	0.761
Graph U-Net	PPCA correction	1	0.654	0.570	0.678	0.658	0.711	0.483	0.672
Transformer	deviations	4	0.663	0.556	0.641	0.718	0.737	0.452	0.658
Transformer	PPCA correction	8	0.648	0.537	0.640	0.697	0.720	0.433	0.639

transformer). The transformer trained with HBCD received more weight next to observed values (0.25–0.34). Refitting the weight table from the validation dumps reproduced the table shipped in DTIPlayground 0.9.8 to two decimals (largest difference 0.00). Cross-fitted, the table was better than the fixed 0.25/0.75 weights with the same transformer (paired 0.987, better on 87%), and 400 training epochs were better than 200 (0.988). Precision weights from the predicted SDs were worse than fixed weights (0.588 vs 0.582; paired 1.012), because the SD attached to a SIREN value is that of the population model, an upper bound next to an observed value; a gradient-boosted weight per cell and calibrated precision weights were about as good as the table (software documentation). Among fixed weights, 0.25/0.75 (0.582) and 0.25/0.5 (0.583) were best; weight 0 near observed values was clearly worse (0.604).

4.6 What did not help

Functa. Over 187 screened runs (configurations and test-time variants), the best functa model reached 0.702 as a hybrid correcting PPCA and 0.736 on the deviations, against 0.720 for PPCA alone and 0.654 for the hybrid U-Net on the same split (Table 4, Fig. 6). The inner gradient steps initially fitted none of a scan's visible values; more steps and test-time refinement helped, but the

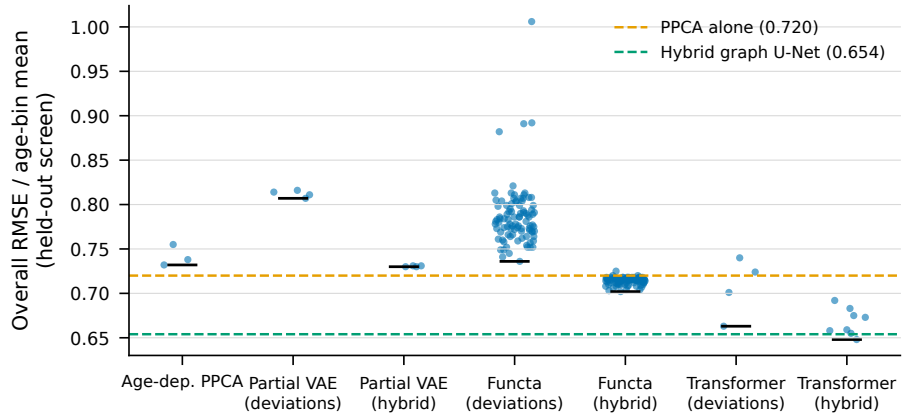


Figure 6: Quick held-out screen: overall error of every screened configuration (dots; bar: best) of the model families that were not adopted and of the transformer, with PPCA alone and the hybrid graph U-Net on the same split.

error then fell only slowly with network size, and the largest networks overfitted the deviations. As a hybrid, functa added little to the PPCA prediction and nothing for regions or a missing family. Graph variants (Laplacian coordinates, a learned vector per location, a graph-convolution decoder) did not help either. A learned vector per location could in principle reproduce the linear PPCA, but fitting the latent by a few gradient steps did much worse than PPCA's exact posterior.

Partial VAE. The best partial VAE reached 0.730 as a hybrid and 0.807 on the deviations, no better than PPCA alone.

Age-dependent PPCA. A population model per age, with kernel weights of log age, was worse than one model of all ages at every bandwidth (0.732–0.755 against 0.720): the deviations from the age-bin norm already remove most of the age effect, and the local models lose scans.

Acquisition correction. BCP scans of the two main acquisitions differed systematically in their deviations (e.g. FA +0.42 SD and ODI −0.27 SD for `shells06select` against `dir79select`; software documentation), but removing these offsets before imputation changed nothing (0.614 vs 0.614; paired 1.001, Holm-adjusted $p = 0.45$). An offset that is constant within a scan is already carried by the scan's own observed values and by the population prediction.

Context inputs and seed ensembles. On seed 0 (54 scans), giving the transformer the distance to the nearest observed value (blend 0.587 vs 0.587; paired 1.001, $p = 0.66$) or the age and acquisition (0.587; paired 1.000, $p = 0.73$) did not help, nor did averaging three transformers trained with different seeds (0.586; paired 0.999, $p = 0.11$).

4.7 A second cohort: pool the training of the networks, not the population model

Pooling HBCD into the norm and the population model did not improve the BCP scans. In the 11-measure phase-3b experiment (Appendix Table A1), pooling with the tensor and NODDI measures of HBCD left the error unchanged (paired 1.010, CI 0.992–1.027, $p = 0.37$), and pooling all measures including the free-water tensor made it worse (1.051, CI 1.031–1.072), because HBCD's free-water fits differed from BCP's by more than a shift and a scale; failed free-water fits with diffusivities far above that of free water had to be removed first. On the 7 measures, pooling the

Table 5: Error relative to the age-bin mean per measure (four patterns, mean over age bins and seeds), and for a whole missing family. ‡First row: absolute RMSE of the age-bin mean (FA, NDI, ODI, FWF $\times 10^3$; MD, RD, AD $\times 10^6$ mm^2/s).

Setting	FA	MD	RD	AD	NDI	ODI	FWF
Age-bin mean: absolute RMSE‡	46.3	102.6	101.6	125.8	65.5	51.8	56.2
Phase 2: SIREN on deviations	0.688	0.638	0.628	0.714	0.613	0.768	0.769
PPCA alone	0.784	0.657	0.652	0.756	0.659	0.818	0.808
PPCA + SIREN	0.663	0.533	0.533	0.639	0.535	0.705	0.689
U-Net alone	0.698	0.604	0.599	0.695	0.605	0.760	0.753
Transformer alone (400)	0.668	0.572	0.570	0.648	0.563	0.699	0.705
Released default (table blend)	0.610	0.504	0.504	0.587	0.505	0.645	0.631
With HBCD reference (table blend)	0.578	0.480	0.478	0.559	0.481	0.610	0.594
Whole family missing (tensor: FA–AD; NODDI: NDI–FWF)							
PPCA alone	0.757	0.648	0.639	0.741	0.758	0.832	0.803
U-Net alone	0.523	0.512	0.502	0.585	0.729	0.680	0.700
Transformer alone (400)	0.490	0.374	0.379	0.456	0.681	0.616	0.573
Released default (table blend)	0.490	0.374	0.379	0.455	0.678	0.615	0.574

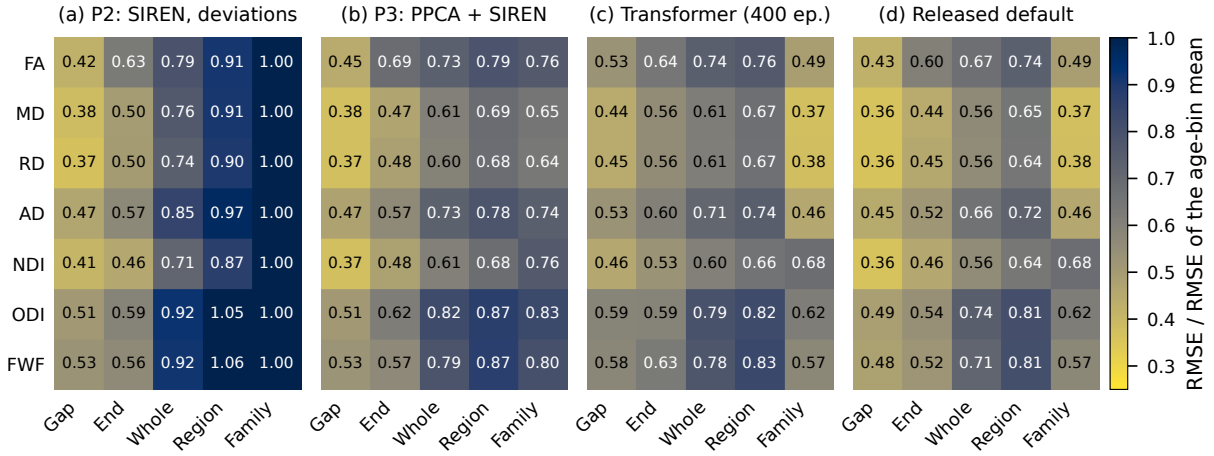


Figure 7: Error relative to the age-bin mean per measure and pattern (mean over age bins and 3 seeds) for four settings. "Family": the measure's family hidden (1.00 for phase 2, which cannot impute it).

norm, PPCA and transformer made PPCA + SIREN worse (0.626 vs 0.614; paired 1.029), mostly for regions (0.833) and a missing NODDI family (0.836).

Training only the transformer on both cohorts, with the norm, PPCA and SIREN of BCP, helped: the transformer alone improved from 0.632 (400 epochs, BCP) to 0.580 (paired 0.923, better on 93%), and the table blend from 0.569 to 0.540 (paired 0.952, better on 92%). The HBCD-trained transformer was not better for a missing NODDI family (0.645 vs 0.622 for the blend).

4.8 Error by measure and age

Measures. The measures fell into two groups (Table 5, Fig. 7). The diffusivities MD and RD and NDI vary smoothly in space and between neighbouring tracts and were imputed best (released default 0.504, 0.504 and 0.505); FA, AD, ODI and FWF depend more on the orientation and organization of a tract and were imputed worst (FA 0.610, AD 0.587, ODI 0.645, FWF 0.631). For ODI and FWF in regions, even the best methods remained near 0.8. For a missing family, the tensor measures were recovered well from NODDI (released default: MD 0.374, RD 0.379, FA 0.490), the NODDI measures less well from the tensor (NDI 0.678, ODI 0.615, FWF 0.574).

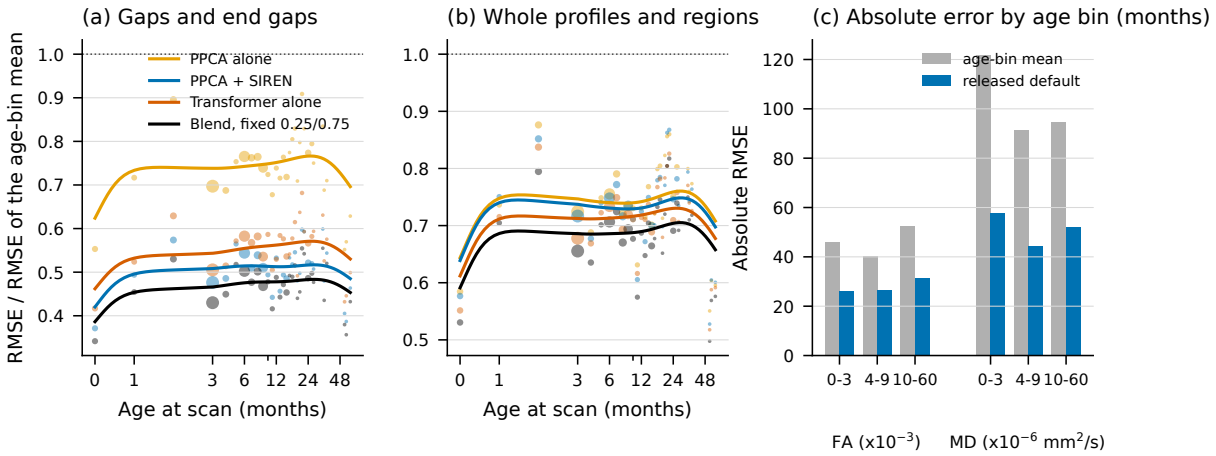


Figure 8: Error by age. (a, b) Error relative to the age-bin mean per age in months (dots, area $\propto$ number of hidden values; lines: Gaussian kernel smooth over log age) for PPCA alone, PPCA + SIREN, the transformer alone and the fixed blend (400 epochs, 3 seeds). (c) Absolute RMSE of FA and MD for the age-bin mean and the released default per age bin.

Table 6: Absolute RMSE (four patterns, mean over patterns and seeds) per age bin; FA, NDI, ODI, FWF $\times 10^3$, MD, RD, AD $\times 10^6$ mm^2/s. Released default and HBCD-reference blend cross-fitted.

Age bin	Setting	FA	MD	RD	AD	NDI	ODI	FWF
0–3 mo	Age-bin mean	46.0	121.9	122.5	138.6	68.1	66.1	63.0
	Released default	26.0	57.9	56.2	80.9	37.0	40.1	41.1
	With HBCD reference	24.2	51.9	50.0	74.1	33.5	36.1	37.1
4–9 mo	Age-bin mean	40.5	91.3	87.5	117.7	58.8	42.0	51.1
	Released default	26.6	44.6	44.2	65.6	30.4	28.1	31.0
	With HBCD reference	25.4	44.3	43.7	64.2	29.8	27.3	30.1
10–60 mo	Age-bin mean	52.4	94.5	94.8	121.1	69.7	47.3	54.5
	Released default	31.4	52.0	51.7	75.3	32.0	30.1	35.0
	With HBCD reference	30.0	50.2	49.7	72.5	31.4	29.0	33.4

Age. Relative to the age-bin mean, the error of the models varied little between age bins (PPCA + SIREN 0.610, 0.610 and 0.622 in 0–3, 4–9 and 10–60 months; fixed transformer blend 0.566, 0.574 and 0.587; Fig. 8a, b), and the ranking of the methods was the same at every age. The per-age curves are noisy where few scans were validated (the youngest and oldest ages). In absolute terms (Table 6, Fig. 8c), the age-bin mean had its largest diffusivity errors in 0–3 months, where values change fastest, and the released default reduced them most there.

4.9 Uncertainty

All model-based methods attach an SD to each imputed value. Calibrated per measure on held-out subjects, the SDs of PPCA alone, the U-Net and the transformer were close to right for whole profiles and regions (root-mean-square of error/SD 0.97, 1.04 and 1.08; 96%, 94% and 94% of errors within 1.96 SD; Appendix Table A2), slightly too small for a missing family (transformer 1.20). The SD of a value imputed by SIREN on the PPCA residuals is that of PPCA, an upper bound: for gaps the errors were only 0.59 of it on average. The SDs of the blend inherited this conservatism near observed values (0.56 for gaps) and were about right for regions (0.98).

Table 7: Paired per-scan comparisons over 162 scan × seed pairs (96 subjects): geometric mean of the ratio of per-scan errors (compared / reference; below 1: compared is better), 95% CI from the pairs and from a subject-level cluster bootstrap, Holm-adjusted Wilcoxon p over the 19 contrasts (on the pairs / on subject means), and the share of pairs where the compared setting is better.

Contrast (compared vs reference)	Ratio	95% CI (pairs)	95% CI (subjects)	p_{Holm} (pairs / subjects)	Better
Population model					
PPCA + SIREN vs PPCA alone	0.834	0.827–0.841	0.827–0.842	$< 10^{-26}$ / $< 10^{-15}$	100%
PPCA + kNN vs PPCA + SIREN	1.082	1.075–1.088	1.074–1.089	$< 10^{-26}$ / $< 10^{-15}$	1%
$q = 40$ vs $q = 80$ (+ SIREN)	1.008	1.002–1.014	1.002–1.014	0.036 / 0.008	42%
$q = 120$ vs $q = 80$ (+ SIREN)	1.010	1.003–1.017	1.002–1.017	$< 10^{-4}$ / $4.3{\times}10^{-4}$	36%
Graph U-Net					
U-Net alone vs PPCA + SIREN	1.085	1.075–1.095	1.074–1.096	$< 10^{-23}$ / $< 10^{-14}$	7%
U-Net beyond 4 mm vs PPCA beyond	0.993	0.990–0.995	0.990–0.996	$< 10^{-7}$ / $< 10^{-5}$	70%
U-Net blend 0.25/0.75 vs PPCA + SIREN	0.966	0.963–0.970	0.962–0.970	$< 10^{-25}$ / $< 10^{-15}$	95%
Transformer					
Transformer alone vs PPCA + SIREN	1.059	1.050–1.069	1.050–1.069	$< 10^{-20}$ / $< 10^{-14}$	13%
Transformer alone vs U-Net alone	0.976	0.970–0.983	0.970–0.983	$< 10^{-9}$ / $< 10^{-6}$	74%
Transformer blend 0.25/0.75 vs PPCA + SIREN	0.947	0.944–0.950	0.943–0.950	$< 10^{-26}$ / $< 10^{-15}$	100%
Transformer blend vs U-Net blend (0.25/0.75)	0.980	0.978–0.982	0.977–0.982	$< 10^{-24}$ / $< 10^{-15}$	94%
Blending					
Precision weights vs fixed 0.25/0.75	1.012	1.010–1.015	1.010–1.015	$< 10^{-12}$ / $< 10^{-9}$	24%
400 vs 200 epochs (fixed blend)	0.988	0.986–0.990	0.986–0.990	$< 10^{-22}$ / $< 10^{-12}$	88%
Weight table vs fixed (400 epochs)	0.987	0.984–0.989	0.984–0.989	$< 10^{-19}$ / $< 10^{-11}$	87%
Released default vs PPCA + SIREN	0.923	0.918–0.928	0.917–0.928	$< 10^{-26}$ / $< 10^{-15}$	99%
Not helpful					
Acquisition correction vs none	1.001	0.999–1.004	0.999–1.004	0.45 / 0.15	48%
Second cohort					
Pooled PPCA + SIREN vs BCP alone	1.029	1.015–1.043	1.014–1.044	$< 10^{-5}$ / $< 10^{-5}$	31%
Reference transformer vs BCP transformer (alone)	0.923	0.913–0.933	0.911–0.934	$< 10^{-22}$ / $< 10^{-13}$	93%
Reference vs BCP transformer (table blend)	0.952	0.945–0.959	0.943–0.960	$< 10^{-22}$ / $< 10^{-12}$	92%

4.10 Statistical summary

Table 7 and Fig. 9 collect the pre-specified paired contrasts. All conclusions held under the subject-level cluster bootstrap and the subject-level Wilcoxon test after Holm adjustment; the bootstrap CIs were nearly identical to the normal-theory CIs, so the dependence between seeds and sessions inflated the precision of the per-pair analysis only slightly. Differences of about 1% (e.g. 40 or 120 vs 80 components) were detectable because all settings saw the same hidden values; their practical relevance is small.

5 Discussion

5.1 Principal findings

The evaluation yields a simple picture. First, *the scan alone does not identify a missing tract*: neighbouring tracts differ systematically, and same-scan methods are only useful for short gaps along a profile. Second, *the normative reference is the main source of information*: imputing deviations from an age-specific norm, rather than values, made all patterns imputable; modelling how deviations co-vary across all tracts and measures (PPCA) improved whole profiles and regions further and made a whole missing measure family imputable. Third, *local and global information are complementary*: SIREN on the PPCA residuals within 4 mm was the best near observed values, the population model and amortized networks beyond. Fourth, *amortized networks that read the observed values directly add to the linear population model*, especially for a missing measure family, but only when they correct the PPCA prediction and are blended with the local method; alone they were worse than PPCA + SIREN for the four patterns. The transformer, with global attention over all tracts, was better than the graph U-Net with its local neighbourhoods. Fifth, *more data*

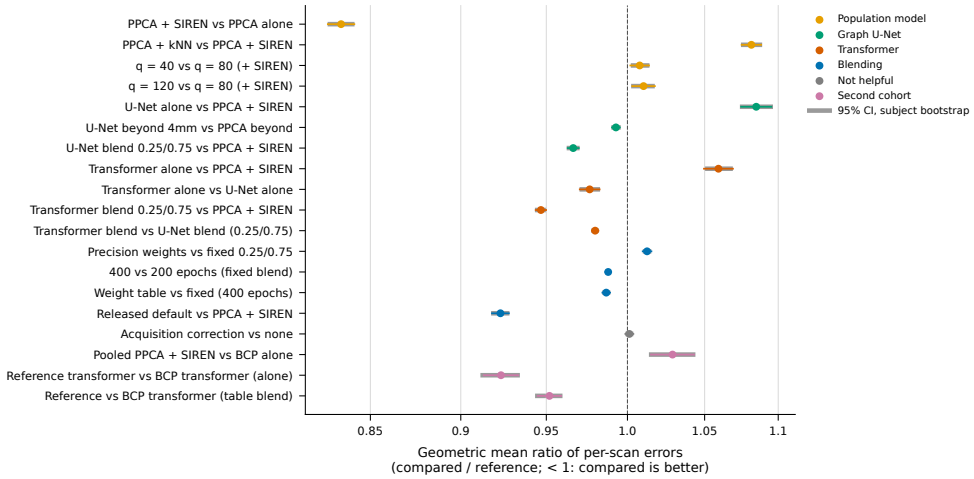


Figure 9: Paired per-scan comparisons of Table 7: geometric mean ratio (dot), 95% CI over scan × seed pairs (coloured line) and subject-level cluster bootstrap CI (grey bar).

helps the networks but not the linear model: training the transformer on a second, larger cohort reduced the error by about 5%, whereas pooling the second cohort into the norm and population model, after an age-aware batch correction, made it worse.

5.2 Interpretation

Why does a linear population model of the deviations work so well, and why does functa fail where the transformer succeeds? The PPCA posterior is the exact conditional expectation under a Gaussian model with one loading per cell; with 80 components it captures the main modes of covariation of 37,716 cells from a few hundred scans. Functa and the partial VAE replace this exact inference by a latent fitted with a few gradient steps or an encoder; with hundreds of scans they did not learn a better prior than PPCA. The graph U-Net and the transformer, in contrast, read each location's observed value directly and only need to learn a correction to the PPCA prediction; this is a much easier learning problem, and their benefit was largest exactly where PPCA has least local information (beyond a few millimetres and for a missing family). The blend weights quantify this division of labour: the transformer received weight 0.16–0.27 next to observed values and 0.57–0.73 beyond 4 mm.

The negative results are informative too. Correcting acquisition offsets did not help imputation because a scan-wide offset is visible in the scan's observed values; this does not mean the offsets are irrelevant for group analyses. Extra context (age, acquisition, distance) did not help the transformer because the observed values already encode them. Averaging seeds did not help because the transformer's variance between seeds is small compared with its bias relative to the truth on held-out subjects.

5.3 Limitations

- *One normative cohort and internal validation.* All methods were developed and compared on BCP, with folds by subject but with the hiding patterns, hyperparameters and defaults chosen on the same three seeds of validation scans. The blend weights were cross-fitted for the reported scores, but the choice among blend variants was made on the same data, so the final numbers may be slightly optimistic. An evaluation on an independent normative cohort is needed [TODO: e.g. HBCD scans as a test cohort, see experiments/requests.md].

- *Synthetic missingness.* Values were hidden completely at random within each pattern. Real missingness is not random: QC rejects tracts in scans with artifacts, more often in some tracts (Fig. 1b) and possibly in some ages or subjects; the true values of rejected profiles may differ systematically from those of similar observed profiles (missing not at random). The validation measures how well a method recovers plausible values, not whether it corrects such biases.
- *Reference of the score.* Errors are relative to the age-bin mean, which favours methods on the age-bin norm less in age bins where the norm is poor (0–3 months, where a continuous norm alone was 13% better). The absolute RMSEs (Table 6) complement the ratios.
- *Sample sizes.* The 0–3 month bin has only 28 BCP scans; 14 of them were validated per seed, and the norm and models of that bin rest on few scans.
- *Dependence.* The 162 scan × seed pairs come from 131 scans of 96 subjects. We addressed this with a subject-level bootstrap and subject-level tests; the screens of extra inputs and members used one seed only, and the functa/VAE screens one split.
- *Measures and atlas.* Results are for 7 measures of one atlas with dense tract axes; sparser tract sets give fewer close neighbours, and the free-water tensor measures were left out after failed fits in HBCD.
- *Downstream impact.* We did not evaluate how imputation affects downstream statistical analyses (e.g. group differences or age trajectories); imputed values are drawn towards the norm where the scan's own information is weak, which can attenuate effects. Imputed values should be flagged (DTIPlayground writes masks and SDs) and sensitivity analyses should exclude them.

5.4 Practical recommendations

1. *Without a normative reference*, fill only short interior gaps (arc-linear along the profile, up to about 15 mm); do not impute whole profiles or regions from the scan's own neighbours.
2. *With a normative reference*, always impute deviations from an age-appropriate norm rather than values.
3. *With a normative cohort of a few hundred scans*, fit a population model (PPCA, about 80 components) of the deviations over all tracts and measures, impute its residuals locally within about 4 mm (SIREN, or kNN without a GPU at about 8% more error), and use the PPCA prediction beyond. This needs no network training and was within 8% of the best setting.
4. *If a GPU is available*, blend this with a transformer trained to correct the PPCA prediction, with weights increasing with the distance to the nearest observed value (the default of DTIPlayground 0.9.8). This is most valuable when whole measure families are missing.
5. *If a larger external cohort on the same tracts exists*, use it to train the transformer (after a batch correction), but keep the norm and population model of the target cohort.
6. *Report* the share of imputed values per tract and measure, keep the imputation masks and SDs, and check that conclusions hold without imputed values. Do not rely on imputed values for group differences between normative and clinical scans, since imputed values are drawn towards the norm.

6 Conclusion

In a developmental cohort with dense atlas tracts, missing diffusion profile values are best imputed as deviations from an age-specific norm, using a joint population model of all tracts and measures, refined locally near observed values and blended with an amortized transformer beyond them. This reduced the error to 0.569 of the age-bin mean's overall (0.540 with a second cohort for training), and to 0.424 and 0.622 when a whole tensor or NODDI fit is missing. Several more complex alternatives did not help. All methods, the validation protocol and the defaults are available in DTIPlayground.

Data and code availability

DTIPlayground (including `dmrifiberprofile impute-profiles` and its `-validate` protocol) is open source at `https://github.com/NIRALUser/DTIPlayground`, on PyPI (`dtiplayground`, version 0.9.8) and on Docker Hub [TODO: Docker image name]. The validation of every phase is documented in the package documentation (`docs/ImputeValidation.rst`). The scripts that compute every number, table and figure of this paper from the validation result files are in `papers/ProfileImputation/scripts` of the repository [TODO: confirm that the paper folder will be public]. BCP data are available through [TODO: NDA collection / BCP data access]; HBCD data through [TODO: HBCD data release / NBDC]. [TODO: Whether derived profile tables can be shared.]

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Author contributions

[TODO: CRediT statement.]

Declaration of competing interests

[TODO: Competing interests.]

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A Supplementary material

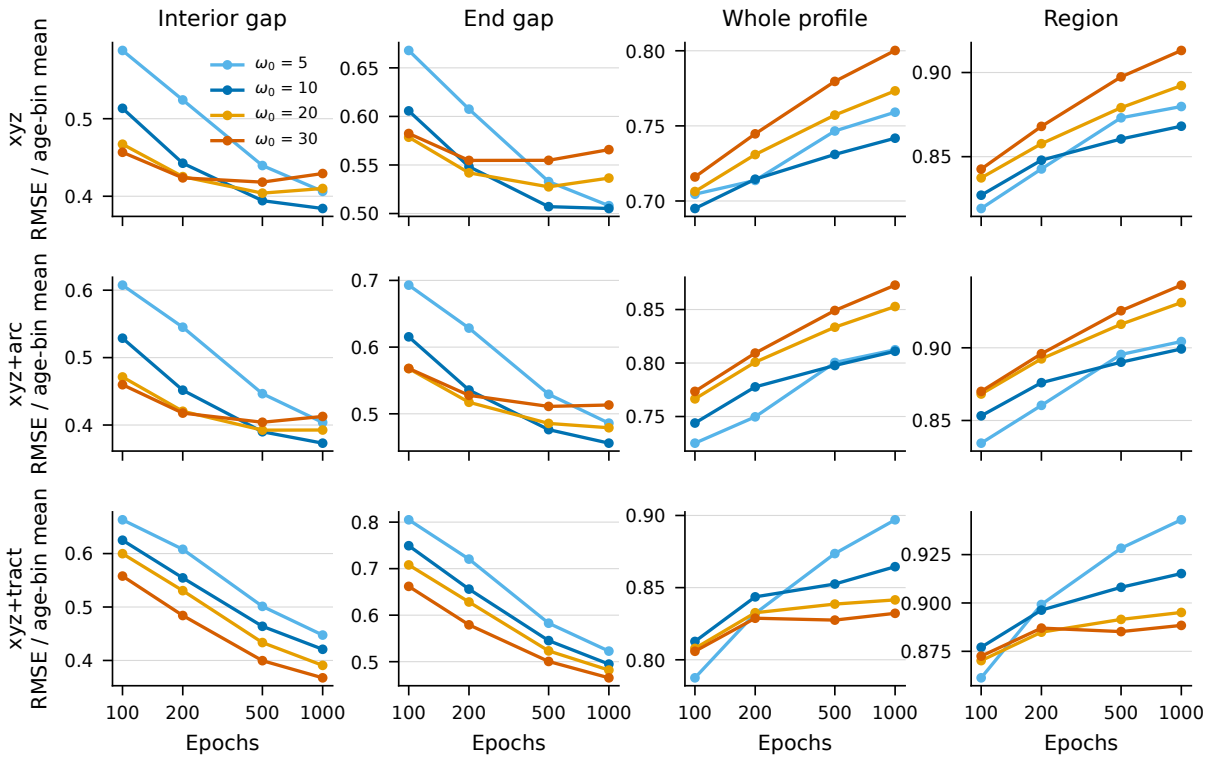


Figure A1: SIREN on the deviations from the age-bin norm: error relative to the age-bin mean against the training epochs, per input (rows), ω_0 (colours) and pattern (columns). Earlier protocol: 30 scans (10 per age bin), one seed, 11 measures (file `docs/_static/impute_validation/phase2/stratified_overall.csv` of the repository).

Table A1: Pooling HBCD and continuous age norms (phase-3b protocol: 3 seeds × 54 BCP scans, 11 measures including the free-water tensor measures; BCP scans scored in BCP units against the BCP-alone age-bin mean). "Pooled tensor+NODDI": HBCD's free-water tensor measures left out. All settings with norm except "Age norm alone" are PPCA + SIREN.

Setting (11 measures, phase-3b protocol)	Overall (SD)	0–3 mo	4–9 mo	10–60 mo
BCP alone, age bins	0.573 (0.007)	0.563	0.567	0.589
BCP alone, kernel age norm	0.584 (0.017)	0.598	0.562	0.593
Pooled tensor+NODDI, age bins	0.564 (0.010)	0.546	0.571	0.577
Pooled tensor+NODDI, kernel age norm	0.620 (0.052)	0.709	0.567	0.584
Pooled all measures, age bins	0.584 (0.010)	0.549	0.599	0.605
Pooled all measures, kernel age norm	0.589 (0.016)	0.561	0.596	0.611
Age norm alone: age-bin mean	1.000 (0.000)	1.000	1.000	1.000
Age norm alone: kernel	0.912 (0.020)	0.868	0.935	0.933
Age norm alone: spline (6/3 df)	0.913 (0.022)	0.869	0.933	0.937

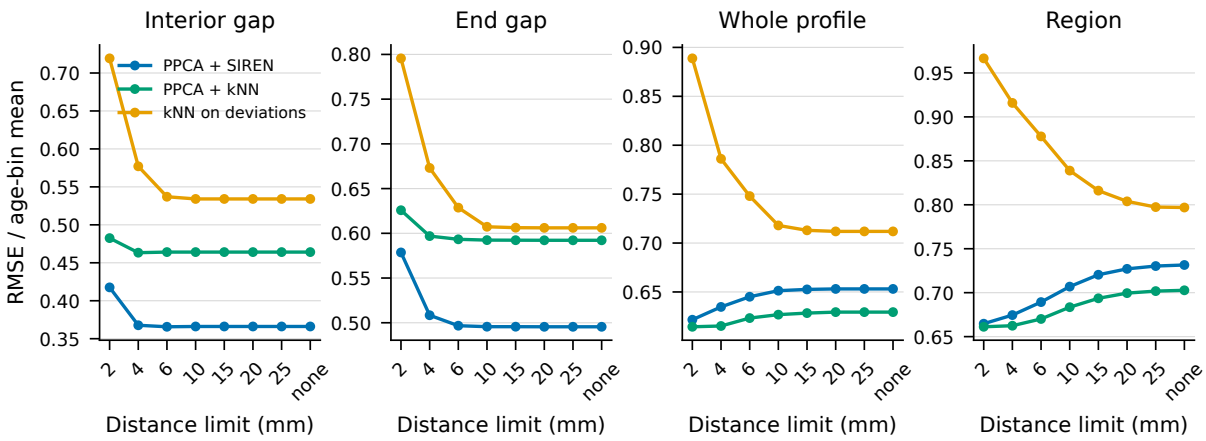


Figure A2: Distance limit of the local methods: error relative to the age-bin mean when the local method fills cells up to the limit and the PPCA prediction (or, for kNN on the deviations, the age-bin mean) beyond. Earlier protocol: 30 scans, one seed, 11 measures (`docs/_static/impute_validation/phase3/grid_caps_long.csv`).

Table A2: Calibration of the predicted SDs: root-mean-square of error/SD (1: right; below 1: SDs too large) and share of errors within 1.96 SD (95%: right), mean over measures, 3 seeds.

Setting	Gap	End	Whole	Region	Family
Age-bin mean (SD of the bin)	1.04 / 95%	1.06 / 94%	1.04 / 95%	1.06 / 94%	1.05 / 94%
PPCA alone (calibrated)	0.92 / 96%	1.09 / 94%	0.97 / 96%	1.06 / 95%	1.06 / 95%
PPCA + SIREN	0.59 / 99%	0.78 / 97%	0.93 / 96%	1.06 / 95%	1.06 / 95%
U-Net alone	0.97 / 95%	1.05 / 94%	1.04 / 94%	1.06 / 94%	1.11 / 94%
Transformer alone (400)	0.97 / 96%	1.02 / 96%	1.08 / 94%	1.08 / 94%	1.20 / 94%
Fixed blend (400)	0.56 / 99%	0.72 / 98%	0.87 / 97%	0.98 / 96%	0.95 / 97%
Precision blend (400)	1.04 / 94%	1.20 / 93%	1.33 / 90%	1.46 / 87%	1.49 / 88%

Table A3: Blend weight tables (weight w of the transformer) refitted from the validation dumps; identical to the tables shipped in DTIPlayground 0.9.8 (`ENSEMBLE_TABLES`).

		SIREN cells (mm)					Fallback cells (mm)				
Table	Measure	0-0.5	0.5-1	1-2	2-3	3-4	4-6	6-10	10-20	>20	Family
BCP	FA	0.27	0.31	0.40	0.51	0.61	0.62	0.65	0.62	0.75	0.98
	MD	0.19	0.24	0.36	0.51	0.60	0.58	0.61	0.70	0.79	0.97
	RD	0.23	0.26	0.36	0.50	0.59	0.57	0.61	0.66	0.85	0.98
	AD	0.23	0.31	0.43	0.56	0.68	0.65	0.71	0.71	0.73	0.96
	NDI	0.16	0.24	0.36	0.49	0.61	0.70	0.68	0.70	0.87	0.82
	ODI	0.25	0.31	0.46	0.58	0.67	0.68	0.70	0.66	0.80	0.96
	FWF	0.20	0.26	0.41	0.47	0.64	0.72	0.70	0.73	0.87	0.93
BCP+HBCD	FA	0.30	0.44	0.57	0.67	0.71	0.69	0.72	0.66	0.69	0.91
	MD	0.31	0.36	0.48	0.62	0.67	0.64	0.61	0.61	0.59	0.87
	RD	0.32	0.38	0.50	0.64	0.69	0.68	0.67	0.64	0.65	0.90
	AD	0.34	0.46	0.57	0.68	0.74	0.68	0.68	0.65	0.59	0.86
	NDI	0.25	0.38	0.51	0.65	0.72	0.69	0.65	0.63	0.65	0.58
	ODI	0.34	0.47	0.61	0.72	0.75	0.71	0.75	0.68	0.66	0.81
	FWF	0.31	0.37	0.46	0.55	0.66	0.56	0.54	0.52	0.59	0.63